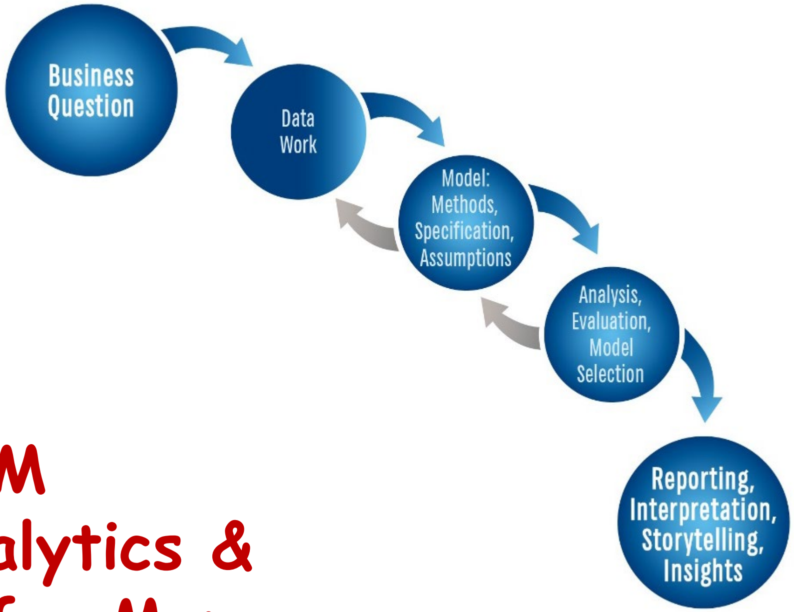


PAML4M
Predictive Analytics & Machine Learning for Managers

6. Machine Learning (ML)

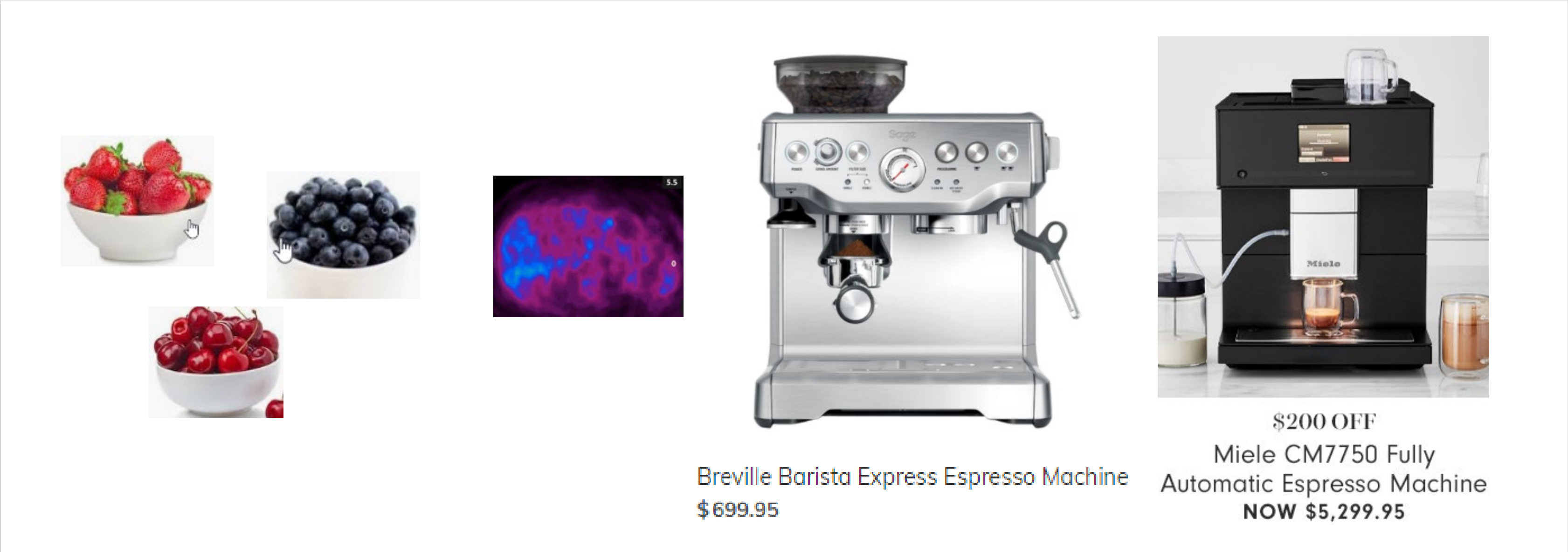
Use PAML4M_6R_MachineLearning.Qmd

Prof. J. Alberto Espinosa

Last updated 8/1/2026

KOGOD SCHOOL *of* BUSINESS
AMERICAN UNIVERSITY • WASHINGTON, DC

Machine Learning (ML) & Cross Validation (CV)



Artificial Intelligence (AI) for Business

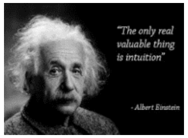
- **AI** is an old discipline – started in 1956 (Darmouth Prof. John McCarthy)
- Defined as: *"the science and engineering of making computers behave emulating human intelligence"* → *learning, reasoning and self-correction.*
  - ➢ *Gartner → AI is the application of advanced analysis, logic-based methods and machine learning (ML) to interpret events and automate decisions.*
- There are many branches of AI, such as:
  - ✓ **Expert Systems** – contain large databases with lots of "if-then" rules elicited from experts → first successful form of AI → composed of a rule (knowledge) base and an inference engine.
  - ✓ **Machine Learning (ML)** – *systems that learn mostly from historical data, without explicit programming* → a branch of AI → predictive AI
  - ✓ **Generative AI (GAI)** – *"deep learning models that can generate high-quality text, images and other content based on data they were trained on"* (IBM) often based on ML and large language models (LLMs), in response to user prompts → process prompt → generate content.

Introduction to ML & CV

- Everything we've done so far is about Machine Learning **(ML)**
- But up to this point, we have focused on model assumptions and good fit and interpretability using the same data set to fit and evaluate the model → e.g., ANOVA F-Tests, R^2, MSE, 2LL, etc,
- We have also evaluated models and predictor variables based on compliance with OLS assumptions → e.g., heteorskedasticity, multi-collinearity, serial correlation, normal distribution, etc.
- **Machine Learning** offers more effective way of doing this, but it focuses is on predictive accuracy, rather than interpretability
- **Cross-Validation (CV)** is central to machine learning
  - ➢ CV is fitting (i.e., training) models with one data set subset and evaluating (i.e., testing) them with a different subset

Benefits of Cross-Validation (CV)

- The increased model variance introduced by multi-collinearity, heteroscedasticity, serial correlation, dimensionality, etc. becomes evident with CV testing.
- **CV** is key !! to evaluate a model's predictive accuracy
- It has two distinct advantages:
  1) Regardless of modeling assumptions/issues, **CV** will always identify the most accurate model and method and specification among the models tested (but it is important to note that the most accurate is not always the most interpretable).
  2) Variable selection methods require models to be nested and of the same type, but **CV** does not and can also compare dissimilar models.



Machine Learning: Intuition

- **ML** is branch of AI that evolved from areas like pattern recognition, computational modeling and artificial intelligence, originally defined as:

  "The field of study that gives computers the ability to learn without being explicitly programmed" (Samuel 1959).

- **ML** is about developing models that can learn from data

- In this class, we view **ML** as: *the development of models that can learn patterns from data (i.e., be trained) and the ability to test these models for predictive accuracy*

Do you need supervision?

Unsupervised Learning

- Machine learning can be supervised or unsupervised.
- We learn many things in life (e.g., how to walk) without even thinking or without specific goals in mind.
- Likewise, when we apply ML methods to learn from the data without a specific goal, this is called "unsupervised" learning.
- **Data mining** is which were previously unknown – i.e., more *"the computational process of discovering trends in data (ACM)"* closely associated with "unsupervised learning".
- When you explore descriptive statistics and correlations you are using unsupervised learning methods.

Supervised Learning

- While we may learn to walk without thinking and without a goal, we often learn other things in life with a specific purpose or goal (e.g., getting a masters degree, learning how to get more customers).
- When we apply machine learning methods with a specific goal in mind, we call this supervised learning
- Most predictive analytics methods fall under the category of supervised learning – there is a specific prediction goal
- The goal of supervised ML is to develop models that predict accurately → not so much about interpretation

Machine Learning Basics

ML Basics

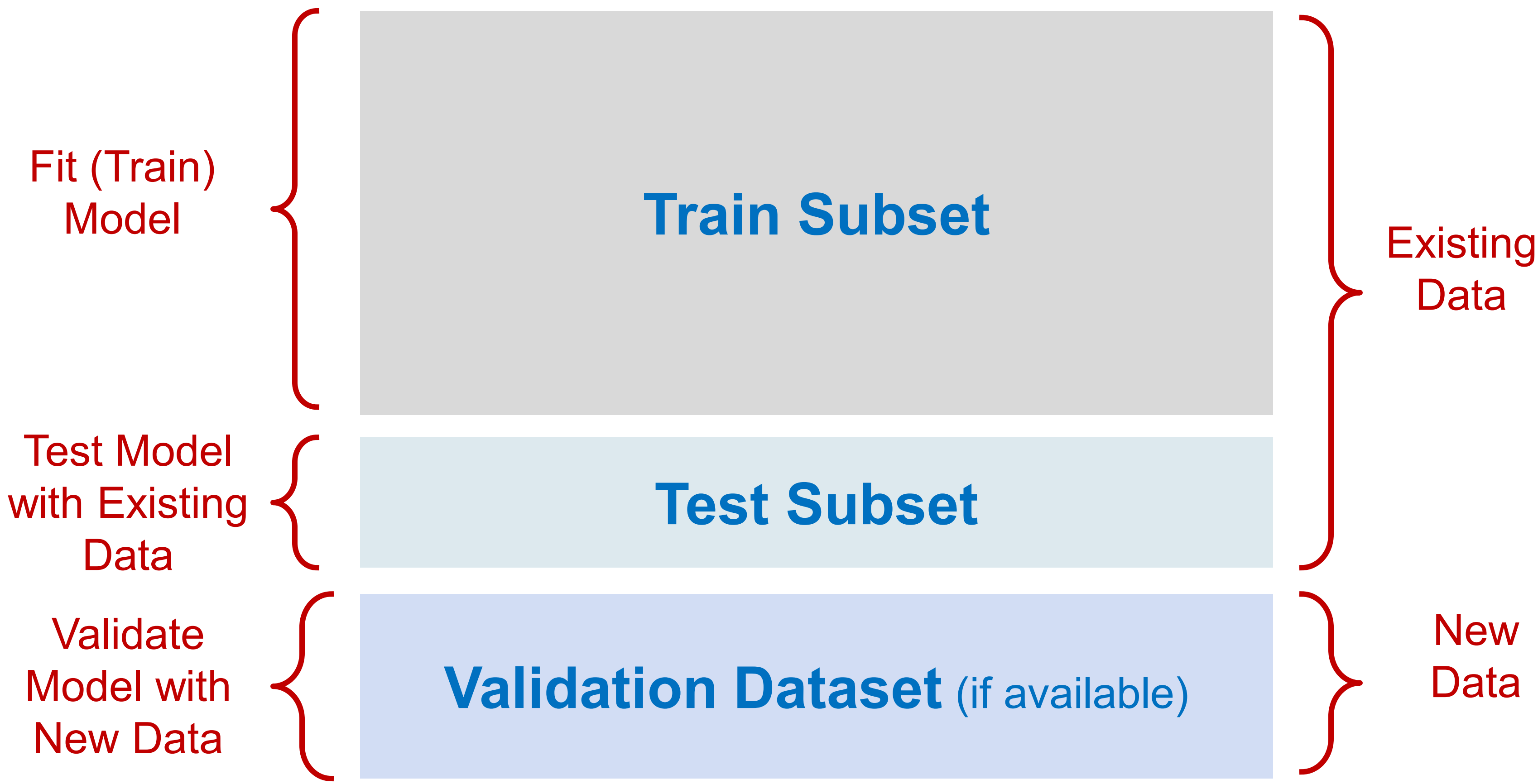
- It helps compare models and model specifications to evaluate the predictive accuracy of each alternative and select the best one
- Using all the data to fit (i.e., train) a model and then using the same data to test the model is **NEVER** a good idea.
- If the model is over-fitted, testing it with the same train data set will be deceptively accurate
- A more common practice is to Train the models with part of the data and then Test them with the remaining data

  → Select the model with the smallest "CV Test Error or Deviance"
- In fact, modern, advanced Deep Learning algorithms (e.g., neural network back propagation) rely heavily on CV to fit models, by making progressive adjustments to the model effects to minimize the Test Error

ML: Key Concepts

- Again, ML is about maximizing predictive accuracy →
  - **Train subset** → part of the data selected randomly to fit models and estimate parameters (e.g., regression coefficients)
  - **Test subset** → part of the data, not in the train subset, used to evaluate the trained models
  - **Validation subset** → is new data (if available) used to further evaluate and validate the models with new data.
  - **Cross-Validation** → Partitioning data into train and test subsets and computing predictive accuracy scores, e.g.
    - CV Test MSE → quantitative models
    - CV Test Deviance → classification models
    - CV Test Classification Error → classification models
    - Etc.

Machine Learning Illustration



Machine Learning:
Illustration: without Cross Validation

Predictors → ***Outcome***

PizzaID	brand	mois	prot	fat	ash	sodium	carb
14001	D	47.17	22.29	21.3	4.08	0.74	5.16
14002	D	49.16	27.99	17.49	3.29	0.39	2.07
14003	A	30.49	21.28	41.65	4.82	1.64	1.76
14004	B	52.68	14.38	25.72	3.26	0.93	3.96
14005	H	33.05	7.34	15.78	1.34	0.42	42.49
14006	H	35.55	7.32	16.4	1.76	0.36	38.97
14007	G	28.68	8.3	16.07	1.41	0.45	45.54

cal
302
278
467
305
341
333
360

Machine Learning:

Cross-Validation → Split, Train, Test, Re-Sample

Training Subset: Train Model → ***Outcome***

PizzaID	brand	mois	prot	fat	ash	sodium	carb	cal
14001	D	47.17	22.29	21.3	4.08	0.74	5.16	302
14002	D	49.16	27.99	17.49	3.29	0.39	2.07	278
14003	A	30.49	21.28	41.65	4.82	1.64	1.76	467
14004	B	52.68	14.38	25.72	3.26	0.93	3.96	305

Testing Subset: Test Model's Accuracy → ***Outcome***

								Actual	Predicted	Error
14005	H	33.05	7.34	15.78	1.34	0.42	42.49	341	290	51
14006	H	35.55	7.32	16.4	1.76	0.36	38.97	333	362	-29
14007	G	28.68	8.3	16.07	1.41	0.45	45.54	360	371	-11

MSE = 1187.7
RMSE = 34.5

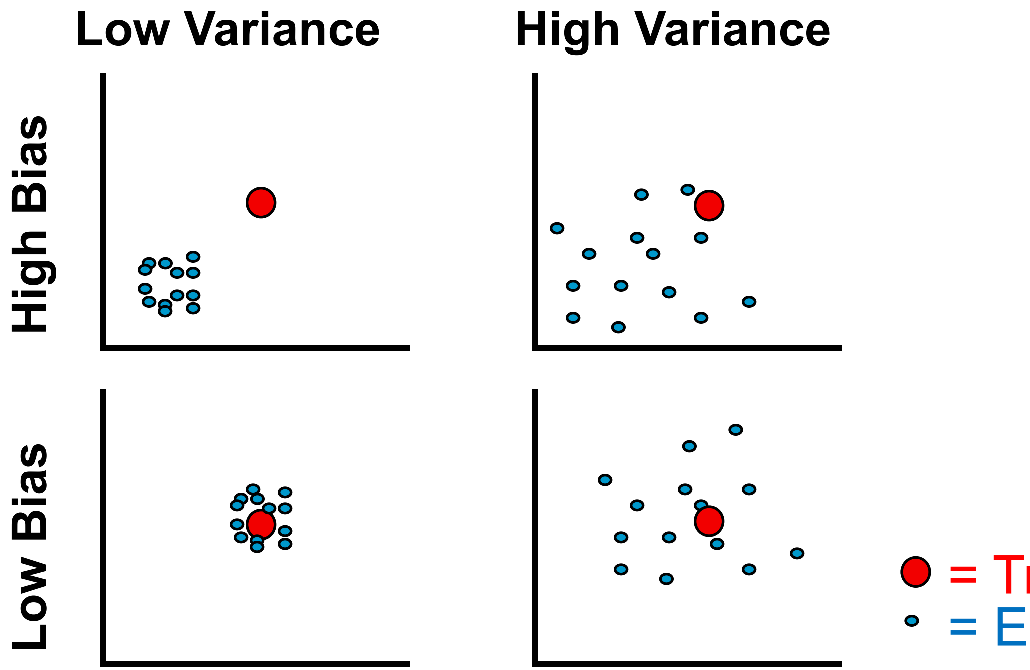
e.g., Cross-Validation (CV) testing, tuning model parameters, deep learning models, Logistic regression, etc.

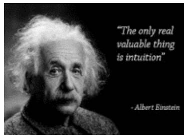
ML: Main Uses of Cross Validation

- Cross-validation (CV) is at the heart of most popular innovations in ML. It can be used for various purposes, including:
  - ✓ Predictive accuracy evaluation → unstable models don't yield consistent results when making predictions with new data. CV helps evaluate the predictive accuracy and stability of models.
  - ✓ Model comparison → similar models (e.g., OLS) are easy to compare, but CV can be used to compare dissimilar models (e.g., OLS vs. Regression Trees) in terms of their predictive accuracy.
  - ✓ Tuning parameter selection → several modeling methods have tuning parameters (e.g., polynomial power, number of variables, number of tree leaves), that analysts can change to tune a model. CV helps evaluate the optimal value of model tuning parameters.
  - ✓ Model estimation → certain methods like neural networks fit models through repeated iterations. CV is used in each iteration to adjust the model recursively and find the best model fit.

Bias vs. Variance Tradeoff

Bias vs. Variance Tradeoff: Intuition





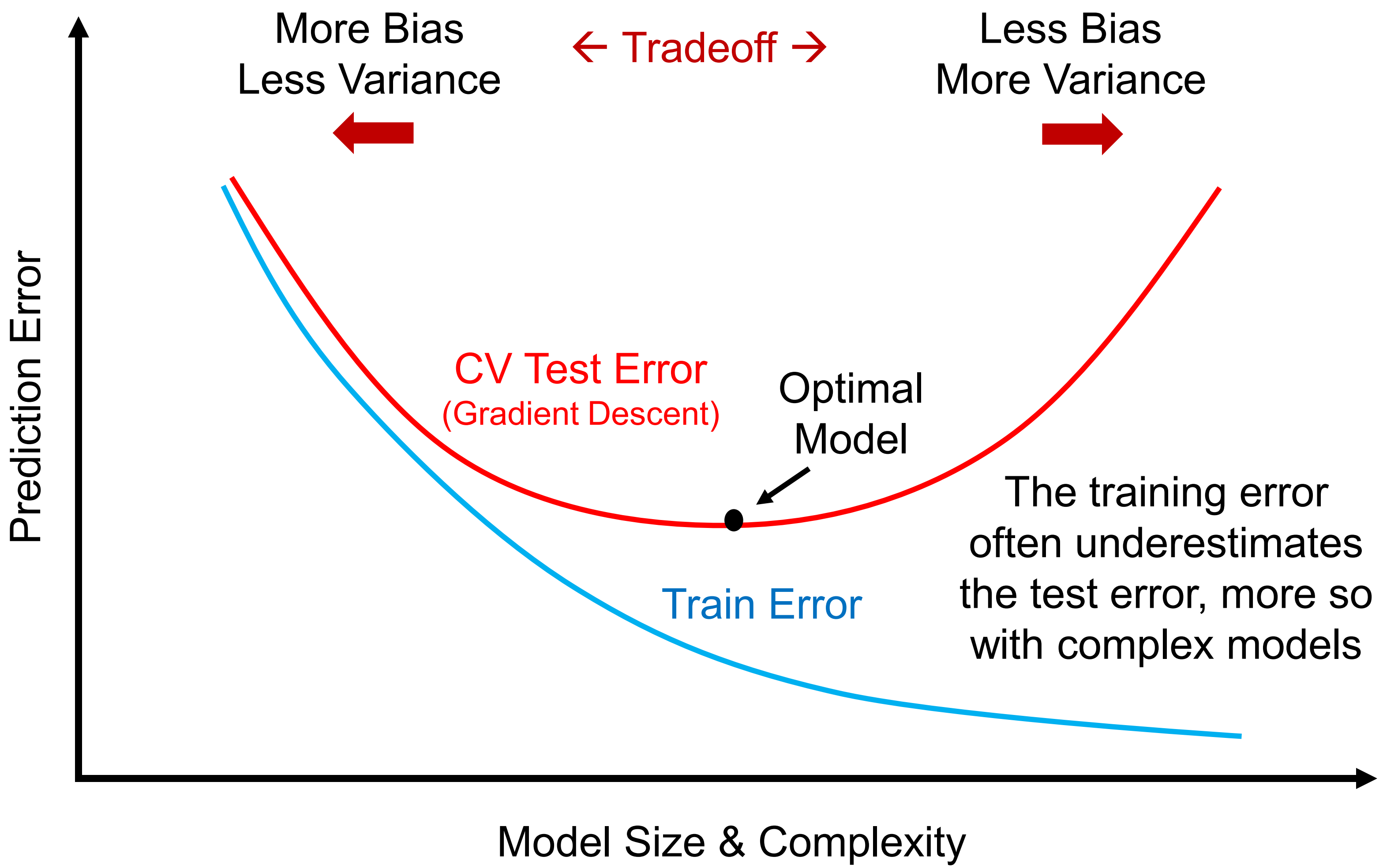
- **Bias** – ability of an estimator or model to predict the correct value
- **Variance** – of the estimator or model across multiple samples
- **Goal** – develop predictive models with low bias and low variance
- This is not always possible → variance vs. bias is a **tradeoff**
- **Low bias** → best for hypothesis testing, inference and interpretation
- **Low variance** → best for predictive accuracy

Train vs. Test Error

- **Train Error** – is the MSE or deviance calculated with the same training data used to develop the model → NOT very useful !!
- When models are over-fitting the training, error tends to be small
- Over-fitted and high variance models are notorious for having high predictive accuracy when evaluated with the same train subset but tend to perform poorly when tested with different data.
- You can test the robustness of your model by dropping a few random data points and see if your model parameter estimates change substantially → "shaking the tree"

- **Test Error** – is the MSE or any deviance statistic calculated with existing data that was NOT used to train the model
- Generally → Test error > Train error → Testing a model with the training subset underestimates the error

Model Complexity & Predictive Error



Error/Fit in Quantitative Models

- $\boldsymbol{MSE}$ → mean squared errors of predicted values → a true measure of error → Low MSE → low $\boldsymbol{RMSE} = \sqrt{\text{MSE}}$ → High R^2 Our focus
- MSE & RMSE ↓ as model complexity and size (p) ↑ → overfitting !!

(Artificial) Adjustments for Model Size and Complexity:

- $Adjusted\ R^2 = 1 - \frac{(1-R^2)(n-1)}{n-p-1}$ → R^2 improves with n; penalized by p
- $Mallow's\ C_p$ is a measure that makes adjustments for model size and complexity:

  $Mallow's\ C_p$ is a constant that adjusts the MSE by applying a penalty for the number of variables "p" (thus the p) and for the variance σ in the errors (same idea than the $Adjusted\ R^2$)

$$Cp = MSE + model\ complexity\ penalty = MSE + \frac{2p\sigma^2}{n}$$

Error/Fit in Classification Models

- **2LL**: maximizing the likelihood of accurate predictions → minimizing the deviance (i.e., deviation of the predictions from actual values) → measured as: $-2 * Log\ Likelihood$

Our focus

- Misclassification Error → proportion of misclassified predictions
- Impurity → How mixed are actual values after classified

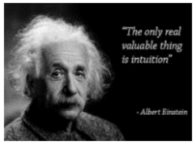
Adjusting for Model Size and Complexity:

- AIC (Akaike Information Criterion) → adjusts $2LL$ for model complexity (k = number of parameters estimated): $AIC = 2LL + 2k$
- BIC (Bayesian Information Criterion) → similar to AIC, but it also factors in the sample size n → $BIC = 2LL + k * log(n)$
- $2LL, AIC\ and\ BIC$ have NO intuitive interpretation with individual models → but are excellent when comparing models → smaller (deviance) is better

Error/Fit Statistics Summary

Model Type	Type of Measure	Raw Statistic	Adjusted for Model Complexity & Other
Quantitative	Explanatory Power	R^2	Adjusted R^2
	Deviance	MSE (or RMSE)	Mallow's CP
Classification	Deviance	2LL	AIC, BIC
	Model Improvement	2LL Null – 2LL Model	AIC/BIC Null – AIC/BIC Model
	Confusion Matrix	Accuracy Rate	Purity (or Impurity)
		Misclassification Rate	Sensitivity (Positives), Specificity (Negatives), False Positives, ROC (and AUC)

Cross-Validation (CV)



Cross-Validation (CV): Intuition

- CV is at the heart of ML
- It provides an excellent evaluation approach for predictive model accuracy, thus its popularity
- CV is particularly important with over-fitted models
- CV is necessary when comparing dissimilar predictive models (e.g., OLS regression vs. regression tree)
- It involves training (i.e., fitting) and testing the model with different data subsets
- CV is NOT a modeling method, but a method to evaluate the predictive accuracy of models

Partitioning and Re-Sampling

- Using only one train/test subset pair is not a good idea because the results will be based on the luck of the draw
- Re-sampling involve drawing random subsets from the data and re-fitting (i.e., re-training) and re-testing the model multiple times
- There are several methods to partition and re-sample subsets.
- Most popular partitioning approaches include:
  - Hold-out or random splitting (set percentage, ex. 70%)
  - K-Fold CV (e.g. 10-Fold) – KFCV (e.g. 10FCV)
  - Leave One Out CV (or Leave P Out) – LOOCV
  - Bootstrapping

Machine Learning & Cross-Validation:

Typical ML process using CV to select the best model:

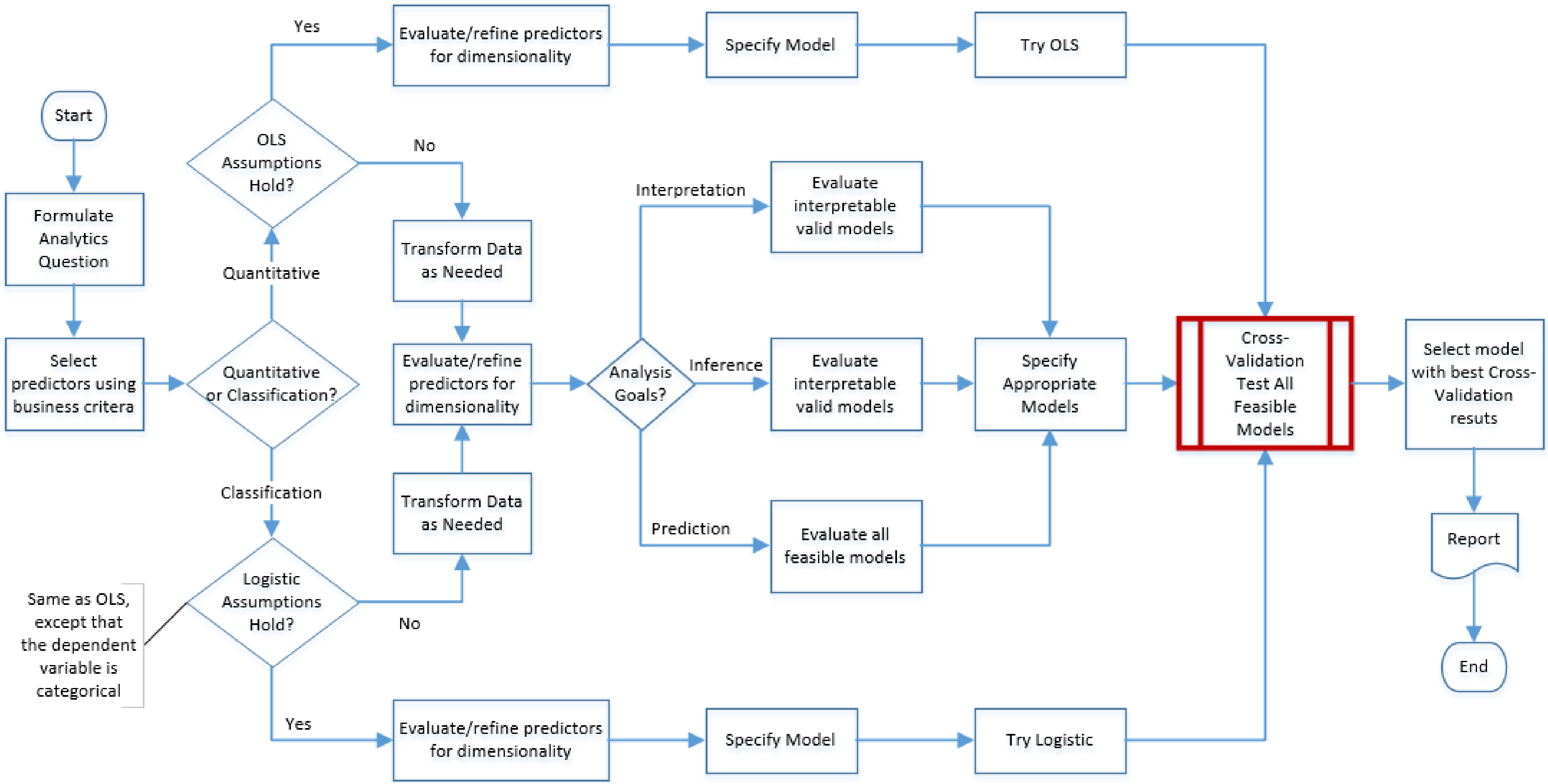
1. Select various models and specifications to evaluate
2. Split the data into train and test subsets
3. Fit (train) the models with the train subset
4. Make predictions with the test subset using the fitted (trained) models
5. Compute the MSE or deviance statistics of the test predictions
6. Re-sample, retrain and re-test the models several times
7. Select the model with the smallest test MSE or deviance
8. Once you select a model, re-fit that model using ALL the data (for statistical power) for actual predictions

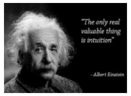
   (most forgotten step !!)

Predictive Model Selection Process

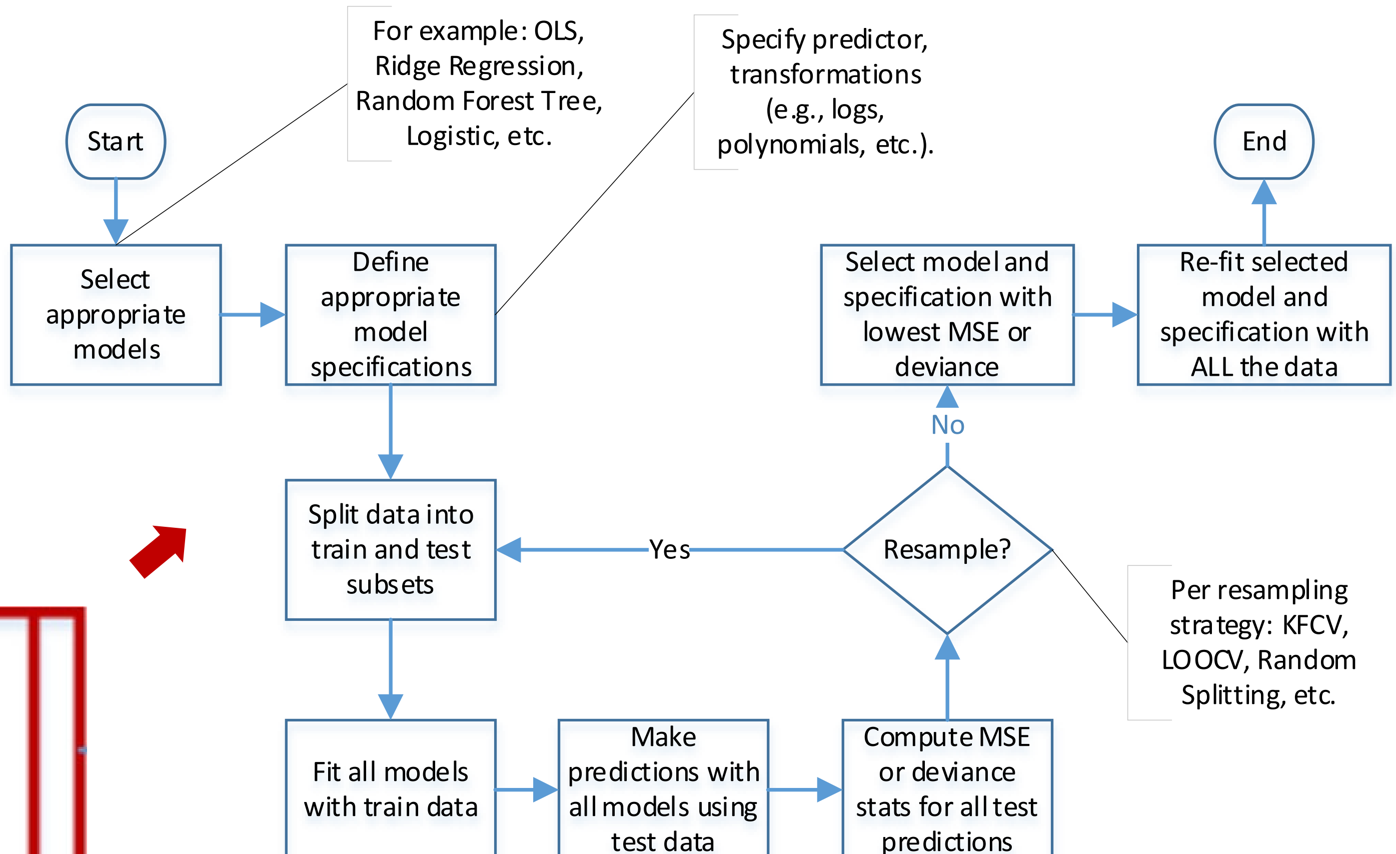
1. Narrow down the model choices based on the response variable type
   - ✓ Try OLS for quantitative predictions first – OLS is BLUE if assumptions hold
   - ✓ Try Logistic for classification predictions first – logit has the same properties as OLS once the response variable is logit-transformed
2. Test OLS/Logistic assumptions → if they hold, you are done
3. If not, make necessary corrections or select appropriate model
4. And/Or:
   - ✓ Narrow the choice of models based on your analytics goals: interpretation, inference (i.e., testing hypotheses), and/or prediction
   - ✓ Do cross-validation (CV) testing – even if OLS assumptions are not fully met, CV testing will find the most accurate model (more on this later).
   - ✓ If CV test results are not very different between models, select the most interpretable model.

Predictive Model Selection Process





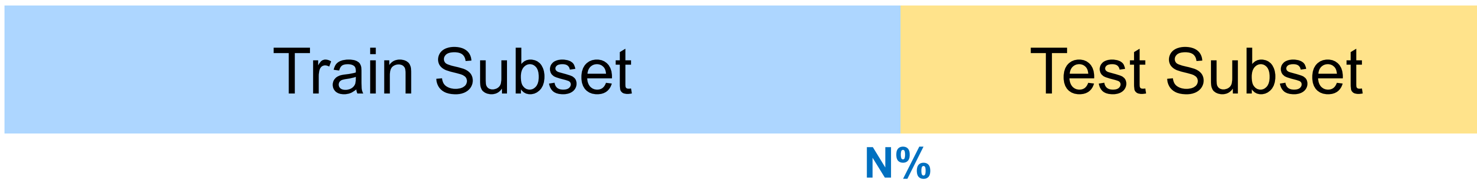
Machine Learning Process



Cross-Validation
Test All Feasible Models

Holdout Sampling or Random Splitting Cross Validation (RSCV)

Holdout/Random Splitting

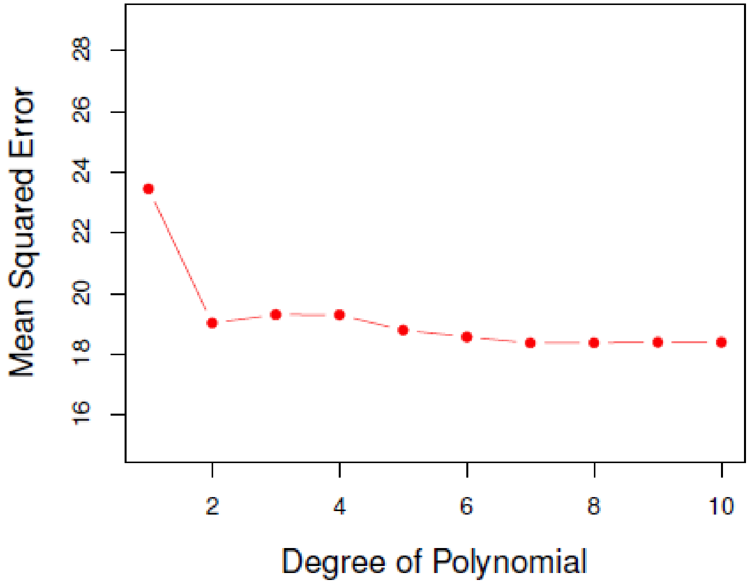


- Randomly select N% of the data for the train subset and hold out the remaining data for the test subset.
- If N is too small (e.g., 50%), then you lose statistical power with the reduced training set, which is problematic with small samples.
- If N is too large (e.g., 90%) the resulting training models may be over-fitted.
- Common values for N are between 70% and 80%
- You can use higher percentages with small data sets and lower percentages with large data sets.
- With a single split, the test MSE may be misleading if you get an "unfortunate" split
- But you can re-split, re-sample, re-train and re-test many times

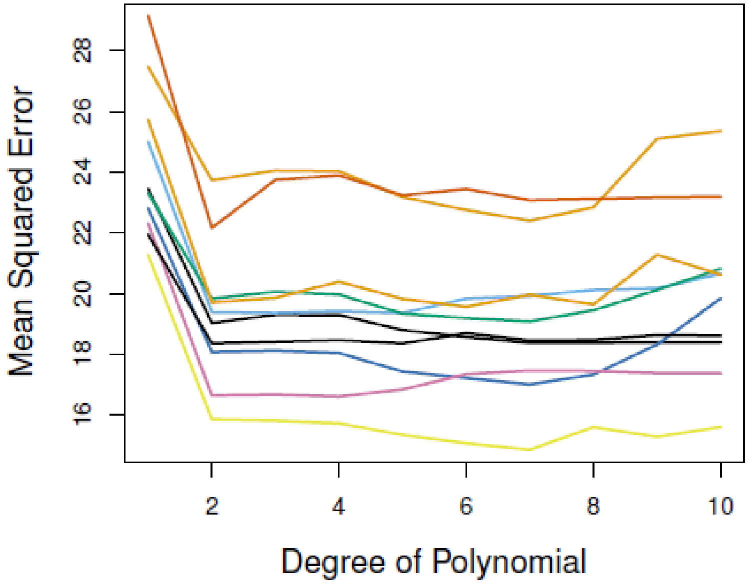
Random Splitting Illustration

The example below was generated with the “Auto” data set in R, predicting gas mileage with horsepower as the predictor using various polynomial regressions. The graph illustrates the variability of the test MSE across splits. All models show that the MSE drops is sharply with a squared regression and that the MSE does not improve substantially with higher polynomials.

Single Split



10 Random Splits

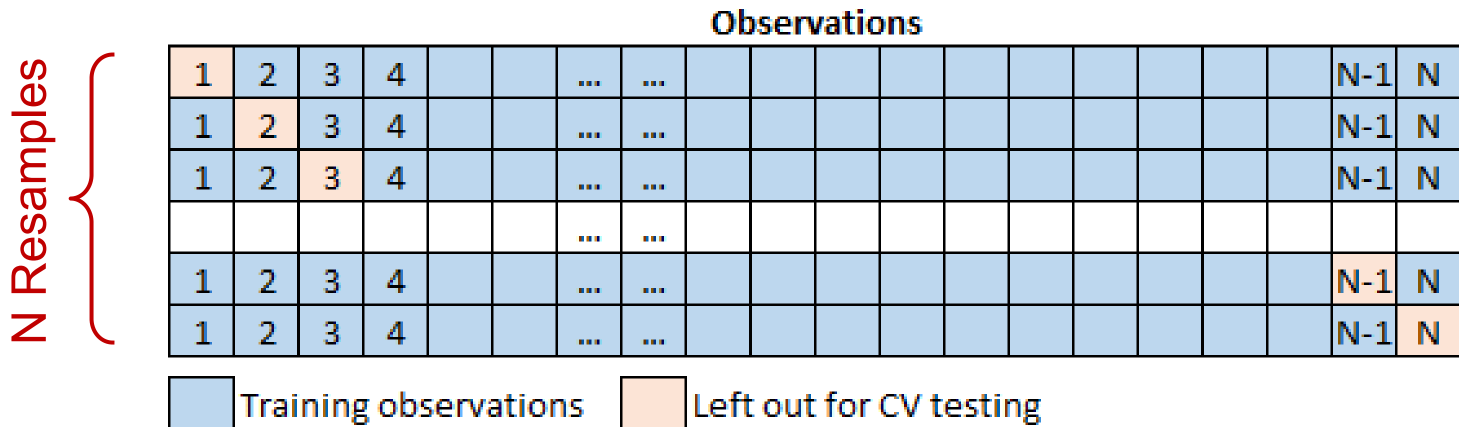


Random Splitting (RSCV)

- {libary} Topic
  `Search keywords`
- Cross-Validation (CV) and Resampling Methods
  `Random Splitting Cross Validation (RSCV)`
  `Random Splitting CV with Re-Sampling`

Leave-One-Out Cross Validation (LOOCV)

Leave One Out Cross Validation



- In LOOCV, the model is trained with N-1 data points and the Test MSE or other fit statistics are calculated with the data point left out.
- Repeated N times, leaving each data point out, one at a time
- This method is computationally intensive – you need to run one regression model estimation for each data point.
- But simulations show that it does NOT outperform the K = 10-Fold cross validation testing (explained shortly).

Leave One Out (LOOCV)

- {libary} Topic
  `Search keywords`
- Cross-Validation (CV) and Resampling Methods
  `Leave-One-Out Cross-Validation (LOOCV)`

K-Fold Cross Validation (KFCV)

K-Fold Cross Validation

K = 10 Resamples

Folds (K = 10 in this example)

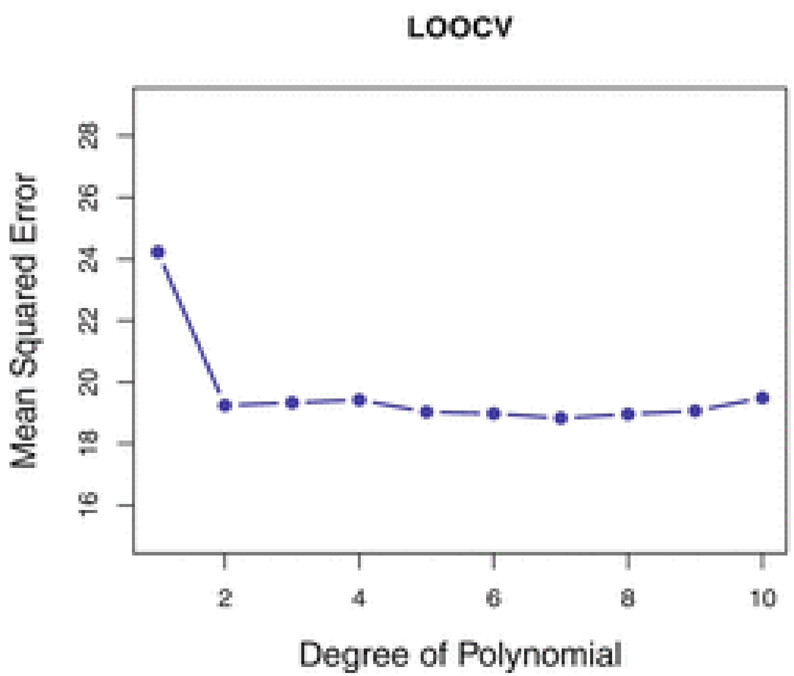
1	2	3	4	5	6	7	8	9	10
1	2	3	4	5	6	7	8	9	10
1	2	3	4	5	6	7	8	9	10
				...	...				
1	2	3	4	5	6	7	8	9	10
1	2	3	4	5	6	7	8	9	10

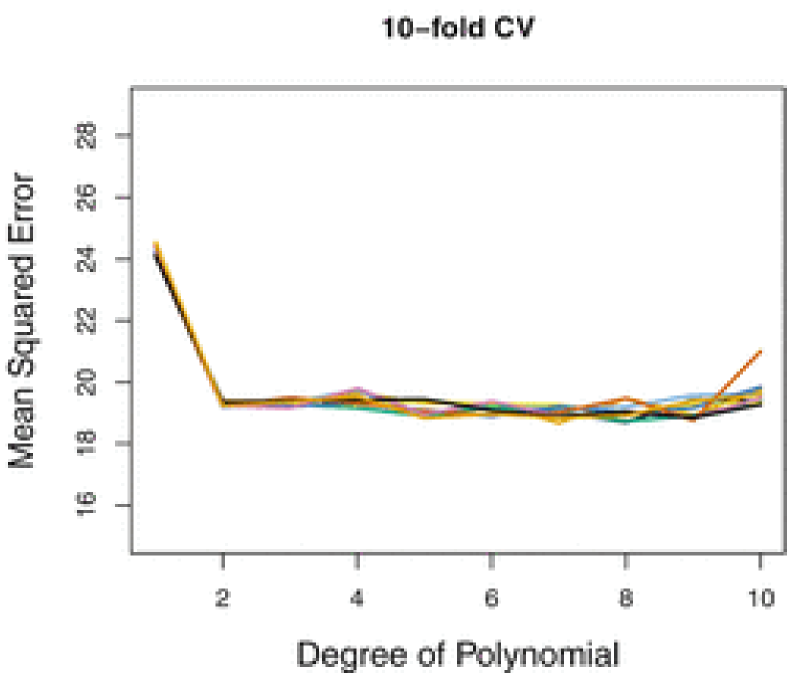
Training folds   Test folds

- **KFCV** is a widely used approach for cross validation testing
- Involves randomly dividing the data into K (roughly) equal parts
- The model is trained on K-1 folds and tested on the remaining fold
- Repeat K times using a different fold for CV testing each time
- The resulting MSE is an average of each of the K MSE calculations
- There is no rule of thumb about the most appropriate value of K
- Smaller K yields smaller train samples → best for large datasets
- Large K is computationally costly → best for smaller datasets
- K = 10 is most popular → train model on 90% of the data each time
- **LOOCV** is identical to **KFCV** when K is equal to the number of observations N (K = N)

LOOCV vs. 10-Fold CV

The example below was generated with the "Auto" data set in R, predicting gas mileage with horsepower (same model we showed before). The two graphs show that the LOOCV and the 10-Fold CV performed similarly, but the 10-Fold CV only requires 10 regression model estimations, whereas LOOCV requires one for each data point.





K-Fold (KFCV)

- {libary} Topic
  `Search keywords`
- Cross-Validation (CV) and Resampling Methods
  `K-Fold Cross-Validation (KFCV)`

Bootstrapping

R Code for Bootstrapping in the {caret} package below

Bootstrapping

- **Bootstrapping** is a popular and conceptually simple statistical approach, applicable to just about any machine learning method.
- Suppose you have a dataset with N observations
- You draw a random sample of N observations with replacement
- Because you sampled with replacement, a few observations may be selected more than once, and some observations will be left out
- This yields a bootstrap sample equal in size to the original data set
- You can then train the model with the bootstrap sample of N observations
- And you can test the model with the observations left out
- You can repeat this process hundreds or thousand of times, no limit

The {caret} R Package

Classification and Regression Training

Overview of the {caret} Package

- This package is fantastic !!. You can do model training (i.e., fitting) with a wide selection of modeling methods, and testing with a wide selection of cross-validation methods, with a single command train(), just by changing a few attributes.
- The {caret} R package was written by the authors of the book "Applied Predictive Modeling" (http://appliedpredictivemodeling.com)
- It has comprehensive documentation at: http://topepo.github.io/caret/index.html
- {caret} is a very useful package when you need to fit many models and test them in different ways, because it uses the same syntax for all model types and cross-validation methods
- So, by simply changing some attributes you can fit a different model and compute cross-validation test statistics.

The {caret} Package

- {libary} Topic
  `Search keywords`
- The {caret} Package
  `The {caret} Package`
- Other {caret} Parameters
  `lm.fit.caret.100 <- train(mpg ~ horsepower + weight +`